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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
CS101 - INTRODUCTION TO PROGRAMMING AND PROBLEM SOLVING

Time: 3 Hrs **Maximum Marks : 60**

Instructions to the candidate:

Figures to the right indicate full marks.
Draw neat sketches and diagram wherever is necessary.
Candidates are required to give their answers in their own words as far as practicable

Part A

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1. Command-line arguments in C are received by which function prototype? **(1)** CO5 BL2 PO1

   (a) int main() (b) void main()
   (c) int main(String args[]) (d) int main(int argc, char* argv[])

2. What will be output of the followingf code? **(1)** CO3 BL2 PO2

```
#include <stdio.h>

void demo() {
   static int x;
   printf("%d ", x);
   x++;
}

int main() {
   demo();   demo();   demo();
   return 0;
}
```

   (a) garbage garbage garbage (b) 0 0 0
   (c) 1 1 1 (d) 0 1 2

3. Which symbol or value is generally used to detect end of file while reading? **(1)** CO5 BL1 PO2

   (a) EOF (b) NULL

(c) 0 (d) \n

4. Dynamic memory allocation in C is performed using: **(1)** CO4 BL2 PO1

(a) newMemry() (b) malloc
(c) free (d) mem_allot()

5. What is the output? **(1)** CO4 BL2 PO1

```
#include <stdio.h>
int main() {
   int a[] = {10, 20, 30, 40};
   int *p = a;
   printf("%d %d\n", *p, *(p + 2));
   return 0;
}
```

(a) 10 20 (b) 10 30
(c) 20 30 (d) 30 40

6. What is the output of the code? **(1)** CO5 BL2 PO1

```
#include <stdio.h>

enum Color { RED, GREEN = 5, BLUE };

int main() {
   enum Color c1 = RED;
   enum Color c2 = BLUE;
   printf("%d %d", c1, c2);
   return 0;
}
```

(a) 0 6 (b) 0 1
(c) -1 6 (d) -1 1

7. Obtain the 2's complement for 11001.11 . **(1)** CO2 BL3 PO3

(a) 00110.01 (b) 00110.00
(c) 11001.00 (d) 00110.11

8. Break statement is used to: **(1)** CO3 BL1 PO3

(a) Exit loop (b) Continue loop
(c) Restart loop (d) End program

9. Which function is used to write a single character to a file? **(1)** CO5 BL2 PO3

(a) fputc() (b) fprintf()
(c) fwrite() (d) putchar()

10. Function to get string length: **(1)** CO4 BL2 PO3

	(a) strcpy() (b) strcmp() (c) strcat() (d) strlen()				
11.	A pointer is (a) a value (b) a variable containing the location of another variable (c) a location (d) None of these	**(1)**	CO4	BL2	PO3
12.	Loop that executes at least once: (a) for (b) while (c) do-while (d) nested loop	**(1)**	CO3	BL2	PO3

Part B
(Answer any four of the following) 5x4=20

13.	Write an algorithm and corresponding flowchart for finding maximum of two numbers.	**(5)**	CO3	BL3	PO1
14.	Write a C program to check whether a number is even or odd.	**(5)**	CO3	BL3	PO4
15.	Write a C program to find factorial of a number using recursion. Explain the concept of recursion.	**(5)**	CO3	BL3	PO3
16.	Answer all				
	(a) What is Dynamic Memory Allocation?	**(1)**	CO4	BL1	PO1
	(b) Write down the differences between malloc() and calloc() function.	**(4)**	CO4	BL2	PO2
17.	(a) Differentiate between break and continue statements.	**(2)**	CO3	BL4	PO2
	(b) Justify the use of break and continue statements in loop with the help of code snippet for each.	**(3)**	CO3	BL5	PO3
18.	Answer all				
	(a) What is structure?	**(2)**	CO4	BL1	PO1
	(b) Write down the differences between structure and union.	**(3)**	CO4	BL4	PO4

Part C
(Answer any three of the following) 3x10=30

19.	(a) Write a C program to Calculate the summation of the following series. Print the summation result.	**(5)**	CO3	BL3	PO3

S= 2+ 4+ 6+ 8 +10 + ……+ up to nth term

(b)	Construct a C Program, with the concept of switch-case, that will accept a character from user and will check whether the taken character is a “Vowel” or “Not a Vowel”.	**(5)**	CO3	BL3	PO3
20. (a)	Write a C program to dynamically allocate memory for an integer array of size n using malloc(). Read n integers from the user, print their sum, and then free the allocated memory.	**(5)**	CO4	BL3	PO3
(b)	Write a C program with two functions, swapByValue(int a, int b) and swapByAddress(int *a, int *b), to swap two integers and demonstrate that call by value does not change the original variables in main(), while call by address (using pointers) successfully swaps their values.	**(5)**	CO4	BL3	PO2

21. Write short note on any ***two***

(a)	Generic pointer	**(5)**	CO4	BL2	PO3
(b)	Computer Memory Hierarchy.	**(5)**	CO1	BL1	PO3
(c)	Bitwise operator	**(5)**	CO2	BL1	PO3
(d)	Entry control and exit control loop	**(5)**	CO3	BL2	PO3

22. Answer the following questions.

(a)	Write a C program to find the length of a string.	**(3)**	CO4	BL4	PO3
(b)	What will be the output of following code snippet?	**(2)**	CO3	BL4	PO3

```
void main( )

{

  int i, j;
 for (i = 1; i <= 3; i++)

 {
   for (j = 1; j <= 3; j++)
```

```
        {
          printf("%d ", i * j);
        }
      printf("\n");
    }

}
```

	(c)	Write a C program to calculate the summation of all element of a 1D array. Take the input as user input.	**(5)**	CO4	BL4	PO3
23.	(a)	Write a C program to copy the contents of one text file into another, where the source file name is "Source.txt" and the destination file name is "Destination.txt". Use appropriate file operations to open, read from the source, write to the destination, and then close both files.	**(5)**	CO5	BL5	PO5
	(b)	Write a C program using command line arguments that accepts two integers and prints their sum. The program should validate that exactly two integer arguments are provided; if not, display a following 3 lines of usage messages. Wrong Input . Proper Useage: ./sum a b e.g., ./sum 10 25	**(5)**	CO5	BL6	PO4